

# HMIC 2019

## March 30, 2019 – April 5, 2019

1. [5] Let  $ABC$  be an acute scalene triangle with incenter  $I$ . Show that the circumcircle of  $BIC$  intersects the Euler line of  $ABC$  in two distinct points.

(Recall that the *Euler line* of a scalene triangle is the line that passes through its circumcenter, centroid, orthocenter, and the nine-point center.)

*Proposed by: Andrew Gu*

Let  $O$  and  $H$  be the circumcenter and orthocenter of  $ABC$ . Recall that

$$\begin{aligned}\angle BOC &= 2\angle A, \\ \angle BHC &= 180^\circ - \angle A, \\ \angle BIC &= 90^\circ + \frac{1}{2}\angle A.\end{aligned}$$

As  $ABC$  is acute,  $A, I, O, H$  all lie on the same side of  $BC$ .

- If  $\angle A > 60^\circ$ , then  $\angle BOC > \angle BIC$ , so  $O$  lies inside  $(BIC)$ .
- If  $\angle A < 60^\circ$ , then  $\angle BHC > \angle BIC$ , so  $H$  lies inside  $(BIC)$ .
- If  $\angle A = 60^\circ$ , then  $O$  and  $H$  are on  $(BIC)$ .

In all cases, two intersections are guaranteed.

*Note:* For an obtuse triangle, the statement is not necessarily true.

2. [7] Annie has a permutation  $(a_1, a_2, \dots, a_{2019})$  of  $S = \{1, 2, \dots, 2019\}$ , and Yannick wants to guess her permutation. With each guess Yannick gives Annie an  $n$ -tuple  $(y_1, y_2, \dots, y_{2019})$  of integers in  $S$ , and then Annie gives the number of indices  $i \in S$  such that  $a_i = y_i$ .

- (a) Show that Yannick can always guess Annie's permutation with at most 1200000 guesses.
- (b) Show that Yannick can always guess Annie's permutation with at most 24000 guesses.

*Proposed by: Yuan Yao*

Part (a) uses the idea that for  $x \neq y$ , the guess  $(x, y, y, \dots, y)$  returns 0 if the first number is  $y$ , 2 if the first number is  $x$ , and 1 if the number is something else. If he tests the pairs  $(x, y) = (1, 2), \dots, (2017, 2018)$ , he will get a 0 or 2 (and therefore find out what the first number is) or get only 1 (so the first number is 2019). In general, with  $n$  numbers unknown, Yannick needs  $\lfloor \frac{n}{2} \rfloor$  guesses to find out the next number in the permutation. Using this strategy, he can guess the permutation within at most

$$\sum_{n=1}^{2019} \left\lfloor \frac{n}{2} \right\rfloor + 1 = 1009 \cdot 1010 + 1 = 1019091$$

guesses.

Part (b) uses the idea that after the first number has been determined and  $n$  numbers remain undetermined, it takes  $\lceil \log_2 n \rceil$  guesses to determine the index of another number, by doing a binary search to halve the candidate positions each time (using the first number as a filler). Using this Yannick can guess the permutation within

$$1009 + \sum_{n=1}^{2018} \lceil \log_2 n \rceil + 1 = 1009 + 1 \cdot 1 + 2 \cdot 2 + 4 \cdot 3 + \dots + 512 \cdot 10 + 994 \cdot 11 + 1 = 21161$$

guesses.

3. [8] Do there exist four points  $P_i = (x_i, y_i) \in \mathbb{R}^2$  ( $1 \leq i \leq 4$ ) on the plane such that:

- for all  $i = 1, 2, 3, 4$ , the inequality  $x_i^4 + y_i^4 \leq x_i^3 + y_i^3$  holds, and
- for all  $i \neq j$ , the distance between  $P_i$  and  $P_j$  is greater than 1?

*Proposed by: Pakawut Jiradilok*

In fact, there are! One might think that the region

$$\{(x, y) \in \mathbb{R}^2 \mid x^4 + y^4 \leq x^3 + y^3\}$$

is inside the ball defined by  $x^2 + y^2 \leq x + y$ , which is a ball of radius  $1/\sqrt{2}$ . It turns out that it is not the case.

We claim that for all  $\epsilon > 0$  small enough, we can choose  $(0, 0)$ ,  $(1, 1)$ ,  $(1 - \epsilon^2, -2\epsilon)$ , and  $(-2\epsilon, 1 - \epsilon^2)$ . First we check the condition  $x_i^4 + y_i^4 \leq x_i^3 + y_i^3$ . This is obviously satisfied for the first two points, and for the other two points this translates to

$$\begin{aligned} (1 - \epsilon^2)^4 + (-2\epsilon)^4 &\leq (1 - \epsilon^2)^3 + (-2\epsilon)^3 \\ (2\epsilon)^3(1 + 2\epsilon) &\leq (1 - \epsilon^2)^3(\epsilon^2) \\ 8\epsilon(1 + 2\epsilon) &\leq (1 - \epsilon^2)^3. \end{aligned}$$

This is satisfied for small  $\epsilon$ . Next we check that the distances are greater than 1. Clearly the third and fourth points are at distance greater than 1 from  $(1, 1)$  as they have negative  $y$  and  $x$  coordinates respectively. They are also at distance greater than 1 from each other for small  $\epsilon$  as their distance tends to  $\sqrt{2}$  as  $\epsilon \rightarrow 0$ . Also,  $(0, 0)$  and  $(1, 1)$  are at distance greater than 1 as well. The only remaining distances to check are the distance from the first point to the third and fourth points (which are the same distance). This distance is

$$\sqrt{(1 - \epsilon^2)^2 + (-2\epsilon)^2} = \sqrt{(1 + \epsilon^2)^2} = 1 + \epsilon^2 > 0$$

so all conditions are satisfied (for sufficiently small  $\epsilon$ ).

For example, one may choose  $\epsilon = 1/20$  to obtain the four points  $(0, 0)$ ,  $(1, 1)$ ,  $(-\frac{1}{10}, \frac{399}{400})$  and  $(\frac{399}{400}, -\frac{1}{10})$ .

4. [10] A *cactus* is a finite simple connected graph where no two cycles share an edge. Show that in a nonempty cactus, there must exist a vertex which is part of at most one cycle.

*Proposed by: Kevin Yang*

Let  $C$  be the original cactus. For every cycle in  $C$ , arbitrarily remove one of its edges, yielding a new graph  $T$ . Observe that since the cycles are edge-disjoint, we removed *exactly* one edge from every cycle, meaning the graph stays connected. However, there are no longer any cycles, so  $T$  is a tree.

Now consider any leaf  $v$  of  $T$  (note that any nonempty tree must have a leaf). If  $v$  is the only vertex in the graph we're trivially done, since then  $v$  was the only vertex in  $C$ . Otherwise  $v$  has degree 1. If  $v$  was originally a leaf of  $C$ , we're done. If not, observe that in the process of turning  $C$  into  $T$ , a vertex's degree cannot decrease by more than half, because for every cycle that a vertex is part of in  $C$ , it gains a degree of 2, but can only lose 1 degree from an edge of that cycle being removed. Therefore, the original degree of  $v$  in  $C$  was at most 2, meaning it could have been part of at most 1 cycle, as desired.

*Remark.* The problem originally did not state that the graph is finite. We apologize for the omission.

5. [12] Let  $p = 2017$  be a prime and  $\mathbb{F}_p$  be the integers modulo  $p$ . A function  $f : \mathbb{Z} \rightarrow \mathbb{F}_p$  is called *good* if there is  $\alpha \in \mathbb{F}_p$  with  $\alpha \not\equiv 0 \pmod{p}$  such that

$$f(x)f(y) = f(x + y) + \alpha^y f(x - y) \pmod{p}$$

for all  $x, y \in \mathbb{Z}$ . How many good functions are there that are periodic with minimal period 2016?

*Proposed by: Ashwin Sah*

**Answer:** 1327392

We will classify all good functions with parameter  $\alpha \in \mathbb{F}_p \setminus \{0\}$ . We will use  $=$  to denote equality modulo  $p$  when appropriate. Let the given statement be  $P(x, y)$ . Then  $P(x, 0)$  gives  $f(x)f(0) = 2f(x)$  for all  $x \in \mathbb{Z}$ .  $f$  cannot be the zero function because it does not have minimal period 2016. Therefore  $f(0) = 2$ .

$P(x, 1)$  gives  $f(x+1) - f(1)f(x) + \alpha f(x-1) = 0$ . This means that  $f$  satisfies a recurrence of depth two.

We can interpret  $f$  as a function  $f : \mathbb{Z} \rightarrow \mathbb{F}_{p^2}$ , embedding the codomain  $\mathbb{F}_p$  inside of  $\mathbb{F}_{p^2}$ . Let  $\tau, \rho$  be the unique roots of  $t^2 - f(1)t + \alpha = 0$  in  $\mathbb{F}_{p^2}$ , so that  $\tau + \rho = f(1), \tau\rho = \alpha$ . Since  $\alpha$  is nonzero, so are  $\tau$  and  $\rho$ . Notice that  $f(0) = \tau^0 + \rho^0$  and  $f(1) = \tau^1 + \rho^1$ . By induction up and down, we see that  $f(n) = \tau^n + \rho^n$  for all  $n \in \mathbb{Z}$ .

We can also easily check that given any value of  $f(1)$  and  $\alpha$  with  $\alpha \not\equiv 0 \pmod{p}$ , we get a unique such function. Thus there are  $p(p-1)$  total such functions, one corresponding to each quadratic  $t^2 - at + b$ , where  $b \neq 0$ . If  $\tau, \rho$  are the roots of the quadratic, the associated function is  $f(n) = \tau^n + \rho^n$ .

Now if  $f$  is  $q$ -periodic then  $f(n) = f(n+q)$  for all  $n \in \mathbb{Z}$ , so  $\tau^n + \rho^n = \tau^{n+q} + \rho^{n+q}$  for all  $n \in \mathbb{Z}$  so

$$\tau^n(1 - \tau^q) = \rho^n(\rho^q - 1).$$

We have two cases.

**Case 1.**  $\tau \neq \rho$ .

Then  $n = 0$  and  $n = 1$  above give  $1 = \tau^q = \rho^q$ , and this is furthermore sufficient to be  $q$ -periodic. Thus the period of  $f$  is the minimal  $q$  such that  $\tau^q = \rho^q = 1$ , so the lcm of the orders of  $\tau, \rho$  as elements of  $\mathbb{F}_{p^2}$ .

**Case 2.**  $\tau = \rho$ .

Then we see that  $\tau = \rho = \frac{f(1)}{2}$ , so they are in  $\mathbb{F}_p$ . This automatically means that  $f(n) = 2 \left( \frac{f(1)}{2} \right)^n$  for all  $n \in \mathbb{Z}$ , where  $f(1) \neq 0$  is forced (as  $\tau\rho = \alpha$ ). This has period dividing  $p-1$ . More specifically, the period is the order of  $\frac{f(1)}{2}$  as an element of  $\mathbb{F}_p$ .

In the first case, note that the orders of  $\tau, \rho$  dividing  $p-1$  implies that  $\tau, \rho$  are actually in  $\mathbb{F}_p$ , so that we can restrict to counting recurrences  $f(n) = a^n + b^n$  where  $a, b$  are nonzero elements of  $\mathbb{F}_p$ .

The unordered pair  $\{a, b\}$  determines  $a + b = f(1)$  and  $ab = \alpha$ , so it determines a unique  $f$  by our earlier observation. We need to count the number of such pairs with  $\text{lcm}(\text{ord}_p(a), \text{ord}_p(b)) = p-1$ .

Recall that for  $d \mid p-1$ , there are  $\phi(d)$  elements of order  $d$  in  $\mathbb{F}_p^\times$ . If  $a = b$  we need  $\text{ord}_p(a) = p-1$ , so there are  $\phi(p-1)$  functions. If  $S$  is the number of ordered pairs  $(a, b)$  with  $\text{lcm}(\text{ord}_p(a), \text{ord}_p(b)) = p-1$ , then we see that the desired count is  $\frac{\phi(p-1)+S}{2}$  (as each unordered pair  $\{a, b\}$  is counted twice as an ordered pair except for those with  $a = b$ ).

Now let  $n = p-1$ . We are counting pairs with the lcm equal to  $n$ . The number of pairs with respective orders  $d_1, d_2$  is  $\phi(d_1)\phi(d_2)$  hence our desired sum is  $\sum_{d_1, d_2 \mid n, \text{lcm}(d_1, d_2) = n} \phi(d_1)\phi(d_2)$ . Now notice that  $\phi$  is a multiplicative function, and the condition on the sum splits among the prime powers dividing  $n$ , so we can choose how  $d_1, d_2$  behave on the prime factors of  $n$  independently. Therefore if we define

$$\chi(n) = \sum_{d_1, d_2 \mid n, \text{lcm}(d_1, d_2) = n} \phi(d_1)\phi(d_2),$$

then  $\chi$  is multiplicative. Finally, if  $q$  is prime and  $e \geq 1$  we have

$$\begin{aligned} \chi(q^e) &= \sum_{d_1, d_2 \mid q^e, \text{lcm}(d_1, d_2) = q^e} \phi(d_1)\phi(d_2) \\ &= 2\phi(q^e)(\phi(1) + \phi(q) + \cdots + \phi(q^{e-1})) + \phi(q^e)^2 \\ &= 2q^{e-1}(q-1)q^{e-1} + q^{2e-2}(q-1)^2 \\ &= q^{2e-2}(q^2 - 1) = q^{2e} \left( 1 - \frac{1}{q^2} \right). \end{aligned}$$

This implies that  $\chi(n) = n^2 \prod_{q|n} \left(1 - \frac{1}{q^2}\right)$  by multiplicativity. Finally, our original answer is

$$\frac{\phi(p-1) + (p-1)^2 \prod_{q|p-1} \left(1 - \frac{1}{q^2}\right)}{2},$$

which we now compute for  $p = 2017$ . We have  $p-1 = 2016 = 2^5 \cdot 3^2 \cdot 7$ . Thus  $\phi(p-1) = 2^4 \cdot 2 \cdot 3 \cdot 6 = 576$  and  $\prod_{q|2016} \left(1 - \frac{1}{q^2}\right) = \frac{3}{4} \cdot \frac{8}{9} \cdot \frac{48}{49} = \frac{32}{49}$ . Thus the number of functions is

$$\frac{576 + \frac{32}{49} \cdot 2016^2}{2} = \frac{576 + 2^{15} \cdot 3^4}{2} = 288 + 2^{14} \cdot 3^4 = \boxed{1327392}.$$